

Lynette Schwanger

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ENVIRONMENTAL SCIENCE & STRATEGIC PROJECT LEADERSHIP SUMMARY

Environmental data scientist with a focus on water resources and carbon management. Integrating a proven background in business, media, and program leadership to bridge science and strategy. Passionate about transforming complex environmental data into actionable solutions that advance climate resilience. My core competencies and skills include:

- Strategic Project Development and Management
- Stakeholder Engagement and Collaboration
- Translating Complex Analyses for Broad Audiences
- Organizational Agility
- Data Management and Analysis
- Environmental Policy Interpretation

WATER RESOURCES AND ENVIRONMENTAL DATA SCIENCE EXPERIENCE

Radical Open Science Syndicate, Fort Collins, CO

March 2026 - Present

Water Database Specialist

- Developed a climate vulnerability indicator for National Park Service water treatment systems to assess resilience to future climate scenarios across 1,900+ water sources
- Built an interactive R Shiny app for NPS staff to visualize and query the Water Supply Database
- Produced geospatial visualizations and data summaries to support NPS water management decisions
- Automated data querying and cleaning tasks using R scripting to improve efficiency and reproducibility

Colorado State University, Fort Collins, CO

Assistant Instructor | Environmental Data Science Applications: Introduction

January 2026 - Present

- Selected as Assistant Instructor after completing the course based on academic performance
- Facilitate weekly office hours providing R troubleshooting and data analysis support to students

Colorado State University - Selected Water Resources Projects

Streamflow Forecasting - Cache la Poudre River

Spring 2026

- Designed and benchmarked six time-series and machine learning models in R to forecast monthly streamflow
- Evaluated models against held-out test data and recommended conservative model for water-supply planning

Forest Carbon Field Data Analysis - Post-Hourglass Fire Vegetation Recovery

Fall 2025

- Conducted transect-based vegetation sampling across five sites to assess post-fire recovery; collected ground cover observations and tree DBH measurements
- Cleaned and organized field datasets and used R to summarize and visualize recovery patterns across sites, including ground cover composition, tree diameter distributions, and species composition

Reproducible Remote Sensing Model - Restored Wetland Photosynthetic Recovery

Spring 2026

- Built and validated a predictive model of wetland gross primary production (GPP) using HLS remote sensing data, spectral indices, and AmeriFlux ground-truth observations
- Applied the model across restored wetland sites on Twitchell Island to assess recovery trajectories and compare seasonal carbon uptake patterns

R Shiny Application - Interactive Fen and Wetland Map

Fall 2025

- Developed an interactive R Shiny application that compares three Colorado wetland and fen datasets and enables polygon-level queries to support transparent, data-informed environmental planning

Colorado State University - Selected Coursework

GIS and Data Analysis in Water Resources

Spring 2026

- Gained proficiency across multiple data platforms used in water resource management, including ArcGIS Pro
- Applied GIS tools to analyze and visualize spatial water resource data, building foundational skills in data ingestion, geoprocessing, and cartographic output

Integrated Ecosystem and Sustainability Analysis

Fall 2025

- Conducted fieldwork and lab work to collect and synthesize data
- Analyzed, interpreted, and reported on water quality and soil carbon data
- Evaluated and implemented sampling strategies

PROJECT MANAGEMENT AND ORGANIZATIONAL LEADERSHIP EXPERIENCE

Lightswitch Video, Leadville, CO (Remote/Hybrid)

CEO | Founder | Partner

January 2019 - May 2025

- Led stakeholder and client meetings, translating technical needs into clear deliverables and timelines
- Fostered strategic partnerships and negotiated contracts to support organizational growth
- Directly grew client account by 250% within four years
- Increased company net worth from \$56,879 to over \$1.8 million

Video Production Manager

May 2013 - January 2019

- Built lasting client relationships with organizations such as Chicago Tribune and Gannett
- Hired and supervised contract teams to deliver high-quality video content for clients
- Improved workflows through continuous process assessment

Video Production Coordinator

November 2011 - May 2013

- Established the Video Operations department, creating efficient workflows to scale operations
- Collaborated across departments to improve digital infrastructure and user experience

EDUCATION

Professional Science Master's, Ecosystem Science and Sustainability

December 2026

Colorado State University, Fort Collins, CO

GPA: 4.0

Bachelor of Arts, English Creative Writing

May 2009

Bachelor of Arts, Cinema

May 2009

Denison University, Granville, OH

GPA: 3.78, *Magna Cum Laude*, Dean's List, Honors Program

VOLUNTEER WORK

Lake County Search and Rescue

May 2024 - Present

- Provide emergency response support, delivering safe and efficient outcomes for those in need
- Apply skills in navigation, risk assessment, and incident command to ensure operational effectiveness in challenging mountain environments

Secretary and Board of Directors Member

January 2026 - Present

- Elected to the Lake County Search and Rescue Board of Directors as Secretary
- Authored a successful \$85,000 grant application to the CO EMTS Funding Program to secure radio equipment
- Track team certifications, attendance, and new member requirements

Blue River Watershed Group - River Watch Sampling

August 2026

- Collected and analyzed water samples in the Blue River watershed for the Colorado River Watch Database

Wildlands Restoration Volunteers - Stream Restoration Volunteer

October 2025

- Built beaver dam analogs and post-assisted log structures to restore floodplain connectivity and wetland habitat